

Theme 6: Vectors

Note Title

5/21/2018

Unit 6.1: Three-dimensional coordinate system. (p. 792)

$\mathbb{R}^1$ - one-dimensional space

$\mathbb{R}^2$ - two-dimensional space

$\mathbb{R}^3$ - three-dimensional space

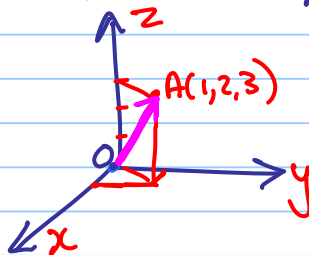
$\mathbb{R}^n$ - n-dimensional space.

Stewart p. 792.

Axes orientation according to the right hand rule:

Example of a point in space:

$A(1, 2, 3)$



$A(1, 2, 3)$ is a point in space.

$$\vec{OA} = \langle 1, 2, 3 \rangle$$

This is the vector to the point $A(1, 2, 3)$

Planes in space:

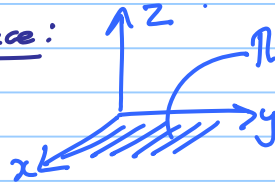
$$z = 0$$

$$y = 0$$

$$y = 1$$

$$z = -1.$$

} see example 1, p. 793.

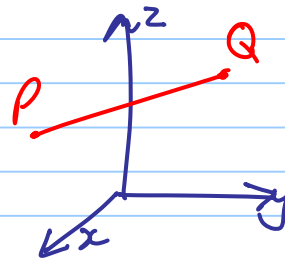


The plane: $z=0$, or: The xy -plane.

Distance between 2 points:

$P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$

MB: Distance formula, p. 795



Examples: (see also example 4, p. 795)

1) Distance between $A(1, 2, -3)$ and $B(4, 0, -3)$

see
p. 795,
p. Exh.

$$\vec{AB} = \langle 3, -2, 0 \rangle$$

(MB: notation)

$$|\vec{AB}| =$$

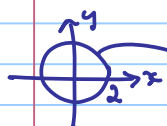
$$= \sqrt{13}$$

2) Distance between $A(2,1,3)$ and the xy -plane.
On the xy -plane: $z=0 \Rightarrow$ we want the distance between
 $A(2,1,3)$ and $B(2,1,0)$ (shortest distance between $A(2,1,3)$
 and the xy -plane.
 $d = |AB| = 3$

3) Distance between $A(3,2,4)$ and the Y -axis.

$\Rightarrow$ we want the distance between $A(3,2,4)$ and $B(0,2,0)$.
 (on the Y -axis: $x=0$ and $z=0$).

The equation of a circle with centre (a,b) and radius r
 is: $(x-a)^2 + (y-b)^2 = r^2$

 Equation of this circle is $(x-0)^2 + (y-0)^2 = 2^2$
 $\Rightarrow x^2 + y^2 = 4$.

Equation of a sphere, centre (a,b,c) , radius r .

See p. 795

Eg. 1) Sphere with centre $(1,2,3)$, radius 4.

$$(x-1)^2 + (y-2)^2 + (z-3)^2 = 4^2$$

2) MB: Example 6 on p. 769.

3) Find the centre and the radius of the
 sphere

$$x^2 + y^2 + z^2 + 2x - 8z + 8 = 0.$$

Here we need to complete the squares.

$$x^2 + y^2 + z^2 + 2x - 8z + 8 = 0$$

$$\underline{x^2 + 2x} + y^2 + \underline{z^2 - 8z} = -8$$

$$\Rightarrow x^2 + 2x + \left(\frac{2}{2}\right)^2 + y^2 + z^2 - 8z + \left(\frac{-8}{2}\right)^2 = -8 + \left(\frac{2}{2}\right)^2 + \left(\frac{-8}{2}\right)^2$$

$$\Rightarrow (x^2 + 2x + 1) + y^2 + (z^2 - 8z + 16) = -8 + 1 + 16$$

$$\Rightarrow (x+1)^2 + y^2 + (z-4)^2 = 9$$

$\Rightarrow$ Centre of sphere: $(-1, 0, 4)$ Radius: 3.

3) Sketch $y=3$ in $\mathbb{R}^2$ and $\mathbb{R}^3$

